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ON LINEAR INTERPOLATION IN THE LATENT SPACE OF DEEP GENERATIVE MODELS

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ABSTRACT

The underlying geometrical structure of the latent space in deep generative models is in most cases not Euclidean, which may lead to biases when comparing interpolation capabilities of two models. Smoothness and plausibility of linear interpolations in latent space are associated with the quality of the underlying generative model. In this paper, we show that not all such interpolations are comparable as they can deviate arbitrarily from the shortest interpolation curve given by the geodesic. This deviation is revealed by computing curve lengths with the pull-back metric of the generative model, finding shorter curves than the straight line between endpoints, and measuring a non-zero relative length improvement on this straight line. This leads to a strategy to compare linear interpolations across two generative models. We also show the effect and importance of choosing an appropriate output space for computing shorter curves. For this computation we derive an extension of the pull-back metric. Code available at: https://github.com/mmichelis/GenerativeLatentSpace

1 INTRODUCTION

Generative models trained in frameworks such as Generative Adversarial Networks (GAN) (Goodfellow et al., 2014) or Variational Autoencoder (VAE) (Kingma & Welling, 2014) have achieved exciting results in computer vision (Karras et al., 2020). In its simplest form, the trained generative model g maps a latent space $\mathcal{Z}$ to some output space $\mathcal{X}$. Vectors populating $\mathcal{Z}$ are sampled according to an arbitrary and fixed latent distribution: $\mathbf{z} \sim \mathbb{P}_Z$, often chosen to be Gaussian. Training, in generative modeling, consists of approximating a target distribution $\mathbb{P}_T$ with the output distribution $g(\mathbf{z}) \sim \mathbb{P}_G$.

$\mathbb{P}_Z$ is distorted by the generator in order to fit the modal characteristics of $\mathbb{P}_T$, creating a highly nonlinear mapping as a result. Hence, navigating in latent space may incur very dissimilar transformations to the output depending on the starting point and the direction picked. Understanding this latent space has been attempted in the form of e.g., finding "meaningful directions" (Voynov & Babenko, 2020; Shen et al., 2020).

Following Arvanitidis et al. (2018), we know that this latent space cannot be regarded as Euclidean without further investigation. Instead, it is equipped with a Riemannian induced metric $\mathbf{M} = \mathbf{J}^T\mathbf{J}$, also called "pull-back metric" (Daouda et al., 2020; Gallot et al., 1993), where $\mathbf{J} = \frac{\partial g}{\partial \boldsymbol{z}}$ is the Jacobian of the generator. The choice of the output space greatly affects the induced metric (Laine, 2018), a choice that we explore in Section 2.2.

We intend to use the previously defined Jacobian to evaluate the average quality of linear interpolations over the whole latent space in the following manner: First, we define the Riemannian curve length, then we sample many shorter curves in latent space and compute the relative length improvement of each quasi-geodesic over the corresponding straight line. The average and standard deviation of these relative improvements are then used as evidence for supporting the discrepancy in linear interpolation quality. Based on these observations, we propose a strategy for comparing interpolations across generative models.

2 METHOD

2.1 SHORTER CURVE

As the generator is highly nonlinear, its associated pull-back metric isn't constant and equal to the identity over the latent space. Hence a straight line is unlikely to be a geodesic, and we can find curves with shorter length that connect two points in latent space. This, in turn, enables a better evaluation of the distance between points. To find shorter curves/geodesics in the Riemannian manifold given by the pull-back metric, we first define the length of a curve $\boldsymbol{\gamma}(t) =: \boldsymbol{\gamma}_t$ (defined for $t \in [0, 1]$ and using shorthand notation $\mathbf{J}_\gamma := \mathbf{J}(\gamma(t))$):

$$\text{Len}(\gamma) = \int_0^1 \|\dot{g}(\boldsymbol{\gamma}_t)\| dt = \int_0^1 \|\mathbf{J}_\gamma \dot{\boldsymbol{\gamma}}_t\| dt = \int_0^1 \sqrt{\dot{\boldsymbol{\gamma}}_t^T \mathbf{J}_\gamma^T \mathbf{J}_\gamma \dot{\boldsymbol{\gamma}}_t} dt \tag{1}$$

Next we choose a method of representing/implementing the curve. Existing methods include e.g., Arvanitidis et al. (2019) using Gaussian Processes, Yang et al. (2018) using quadratic functions, or Laine (2018) using a discrete array of points. We opted for a continuous curve with analytical derivatives, where control points only change behavior locally: B-splines. As we need at most second order derivatives, we chose cubic B-splines (see implementation details in Appendix A).

With this curve implementation, we can either directly minimize $\text{Len}(\gamma)$ via optimization, or solve the geodesic Ordinary Differential Equation (ODE) defined and derived in Arvanitidis et al. (2018) for finding the minimizer γ of Equation 1. The ODE requires a second derivative i.e., the Hessian, which is computationally much more expensive, and furthermore did not consistently find shorter curves than the direct length minimization approach in our experiment. The ODE approach does, however, provide a measure of convergence, i.e. how close the solution is to the geodesic. In the end, we chose not to use the ODE, and instead minimize Equation 1 for finding shorter curves in latent space by optimizing the control points of the cubic B-spline using gradient descent. To improve the convexity of $\text{Len}(\gamma)$ and have a better behaved optimization, we instead minimize $\int_0^1 \|\dot{g}(\boldsymbol{\gamma}_t)\|^2 dt$ i.e., the Path Energy, which does not change the minimizer of the former functional.

We initialize the cubic B-spline with a straight line with fixed start and end points plus two variable control points in between. Whenever the curve length plateaus, we add a new control point and resume optimization. Termination criteria are maximal node count and number of optimization steps. The drawback of this method is that we cannot claim that the result is a geodesic, it is simply a shorter curve than the straight line. Consequently, its length provides a more accurate "distance" between points than measuring the straight line's Riemannian length.

2.2 JACOBIAN

The pull-back metric depends solely on the Jacobian of the generator, which requires special care when being computed. For a deterministic generator, it is enough to backpropagate the derivatives from the output of the generator to the inputs. In the case of a stochastic generator/decoder as in VAE, we use the expected value of the induced metric, combining the Jacobian for the mean and standard deviation of the outputs (Arvanitidis et al., 2018):

$$\mathbf{M}_z := \overline{\mathbf{M}}_z = (\mathbf{J}_z^\mu)^T \mathbf{J}_z^\mu + (\mathbf{J}_z^\sigma)^T \mathbf{J}_z^\sigma \tag{2}$$

An open question is whether the output space $\mathcal{X}$ of the generator (which is often an image) is meaningful for the metric computation, hence it is worth investigating what the effect is of "feature mappings" $f : \mathcal{X} \to \mathcal{F}$ on top of the generator. Two options we implemented were the logistic regression output and activations of a VGG-19 network. For the former, a clear and intuitive effect can be observed when tested for MNIST digits: geodesics pass through as few digit clusters as possible in latent space (see Appendix B). VGG activations are assumed to structure the latent space in a perceptually meaningful way: Laine (2018) investigates this claim qualitatively, while Moor et al. (2020) does so quantitatively.

For further (deterministic) mappings on top of a deterministic generator, we simply multiply the Jacobians to compute the overall induced metric:

$$\mathbf{M}_{FZ} := \mathbf{J}_{XZ}^T \mathbf{M}_{FX} \mathbf{J}_{XZ} \tag{3}$$

Here $\mathbf{J}_{XZ}$ stands for the Jacobian of the generator $g : \mathcal{Z} \to \mathcal{X}$, and $\mathbf{J}_{FX}$ would be for the feature mapping $f : \mathcal{X} \to \mathcal{F}$, with $\mathbf{M}_{FX} := \mathbf{J}_{FX}^T \mathbf{J}_{FX}$. In the case of a stochastic generator (under assumption of diagonal covariances):

$$\mathbf{M}_{FZ} := \overline{\mathbf{M}}_{FZ} = \left(\mathbf{J}_{XZ}^{\mu}\right)^T \mathbf{M}_{FX} \mathbf{J}_{XZ}^{\mu} + \left(\mathbf{J}_{XZ}^{\sigma}\right)^T \hat{\mathbf{M}}_{FX} \mathbf{J}_{XZ}^{\sigma} \tag{4}$$

Here $\hat{\mathbf{M}}_{FX}$ keeps just the diagonal entries of $\mathbf{M}_{FX}$, i.e., all off-diagonals are set to 0. We only need the diagonals for variance as we assume diagonal covariances for the normal distribution in output space (e.g. VAE). The exact derivation of this result can be found in Appendix E.

2.3 Evaluation of Linear-to-Geodesic Deviation

To quantify how much the linear interpolations deviate from the geodesics on average, we define and measure the expected worst-case relative improvement of the Riemannian length between pairs of points by sampling over the whole latent space. We can sample a starting point from the latent distribution $\mathbb{P}_Z$, and move in the direction of the eigenvector with the largest eigenvalue of the pull-back metric at that point; a direction we call maximal eigenvector for short. For a given step-size, we now have a start and end point, and can compute the relative improvement of a shorter curve compared to the Riemannian length of the straight line connecting both points. Lastly, we take an average of this value over all the samples in latent space. The process can be described as in Algorithm 1 in Appendix C.

The result on a VAE can also be found in Appendix C. We follow the maximal eigenvectors at a given point in the latent space[1] to induce maximal change in output, which computes the "worst-case" and enables us to obtain an upper bound on how much the worst straight lines can be improved. We found that the metric is highly anisotropic (see the large condition numbers in Figure 1), and while Wang & Ponce (2021) observed that the maximal eigenvectors are similar at different positions in the latent space, we remark that for our experiments they were not homogeneous, but varied in direction throughout latent space (see Figure 1).

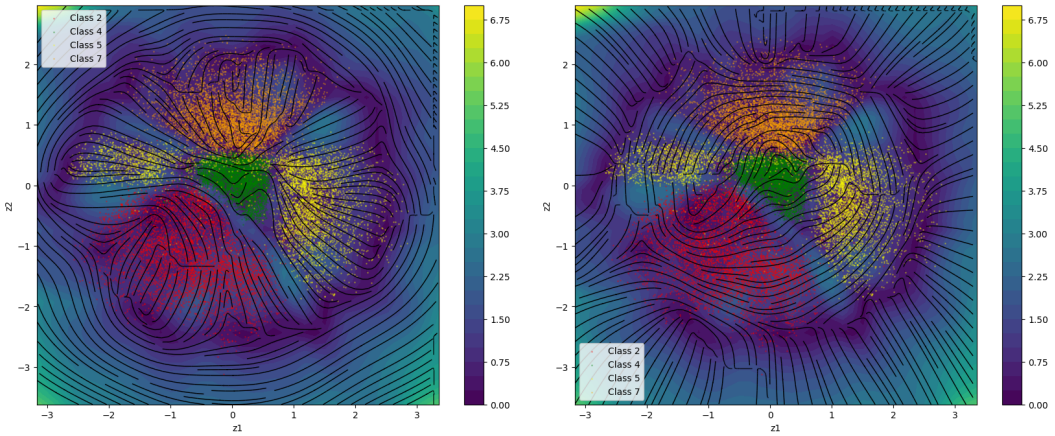

Figure 1: The latent space of this VAE is 2D. In the background the logarithmic condition number i.e., ratio of largest to smallest eigenvalue is plotted, together with clusters of encoded MNIST test data (digits 2,4,5,7). In the foreground we see streamlines following the minimal eigenvectors i.e., eigenvectors of smallest eigenvalue of pull-back metric at every point on the left, and maximal eigenvectors on the right. We used the improved VAE variance estimate of Arvanitidis et al. (2018).

[1]We aligned the maximal eigenvectors such that it always points towards the origin, this prevents the end point to lie in regions where the generator was not trained for